

ArraySim5-3: Site-Practice Heterogeneity Data-Generating Process, Oracle Truth, and CCW Estimators

1 Purpose and simulation structure

ArraySim5-3 evaluates clone-censor-weight (CCW) estimation when treatment initiation practices differ systematically across sites even after adjustment for measured baseline characteristics and time-varying clinical history. The simulation deliberately holds patient mix constant across sites. Therefore, between-site differences in treatment initiation are generated by a residual site-practice effect rather than by differences in the distribution of measured patient characteristics.

There are three sites, indexed by $k \in \{1, 2, 3\}$, with 1,000 patients per site. Patients are indexed by i , and discrete follow-up intervals are indexed by $m = 1, \dots, M$, where $M = 25$. The simulation grid is

$$\tau \in \{5, 10, 15\}, \quad \beta_{\text{trt}} \in \{-1.0, -0.7, -0.5\}, \quad h \in \{\text{low, moderate, high}\},$$

where τ is the grace-period length and h indexes the magnitude of site-practice heterogeneity. This gives $3 \times 3 \times 3 = 27$ simulation cells. Each cell contains 100 Monte Carlo replicates.

2 Notation

For patient i at site k , let

- X_{1ik} and X_{2ik} denote baseline covariates;
- L_{1ikm} and L_{2ikm} denote time-varying covariates;
- S_{ik} denote the interval of treatment initiation;
- T_{ik} denote the terminal-event interval;
- $A_{ik}^\tau = \mathbb{I}(S_{ik} \leq \tau)$ indicate initiation within the grace period; and
- $g \in \{1, 0\}$ index the strategies “initiate within τ ” and “do not initiate within τ ,” respectively.

If a patient does not initiate or experience the event during observed follow-up, the corresponding time is coded as $M + 1$.

3 Data-generating process

3.1 Baseline patient mix

The baseline distributions are identical at all three sites:

$$X_{1ik} \sim N(0, 1), \quad X_{2ik} \sim \text{Bernoulli}(0.40).$$

Thus,

$$(X_{1ik}, X_{2ik}) \mid K_i = k \stackrel{d}{=} (X_{1ik'}, X_{2ik'}) \mid K_i = k'$$

for every pair of sites k and k' . Site-practice heterogeneity is therefore not induced by different measured patient populations.

3.2 Time-varying covariates

For $\ell \in \{1, 2\}$, the time-varying covariates follow

$$L_{\ell,ikm} = 0.6L_{\ell,ik,m-1} + 0.3X_{1ik} + 0.2X_{2ik} + \varepsilon_{\ell,ikm}, \quad \varepsilon_{\ell,ikm} \sim N(0, 1),$$

with $L_{\ell,ik0} = 0$. Innovations are generated separately for L_1 and L_2 . Treatment does not affect the subsequent covariate process. Covariates are generated only while a patient remains alive; values after a terminal event are left undefined.

3.3 Treatment initiation

Among patients who are alive and have not initiated before interval m , let

$$q_{ikm} = \Pr(S_{ik} = m \mid S_{ik} \geq m, T_{ik} \geq m, X_{ik}, L_{ikm}, K_i = k).$$

The treatment-initiation DGP is

$$\text{logit}(q_{ikm}) = \alpha_{kh} + 0.8X_{1ik} - 0.3X_{2ik} + 0.5L_{1ikm} + 0.4L_{2ikm}. \quad (1)$$

The scenario-specific site intercepts are

Table 1: Site-specific treatment-initiation intercepts.				
Practice heterogeneity	Site 1	Site 2	Site 3	Range
Low	−3.25	−3.00	−2.75	0.50
Moderate	−3.75	−3.00	−2.25	1.50
High	−4.50	−3.00	−1.50	3.00

Every scenario is centered at -3 . Consequently, the scenario axis changes the degree of between-site practice heterogeneity without uniformly shifting the treatment-initiation intercept across all sites. At zero covariate values, the corresponding per-interval probabilities are approximately

Practice heterogeneity	Site 1	Site 2	Site 3
Low	3.7%	4.7%	6.0%
Moderate	2.3%	4.7%	9.5%
High	1.1%	4.7%	18.2%

Thus, two patients with the same measured history may have different initiation probabilities solely because they receive care at different sites.

3.4 Terminal-event process

Conditional on being alive at the start of interval m , define the event hazard

$$p_{ikm}^Y = \Pr(T_{ik} = m \mid T_{ik} \geq m, \mathcal{H}_{ikm}),$$

where $\mathcal{H}_{ikm}$ contains the observed covariate and treatment history. The event DGP is

$$\begin{aligned} \text{logit}(p_{ikm}^Y) = & -2.5 + 0.5X_{1ik} + 0.4X_{2ik} + 0.4L_{1ikm} + 0.3L_{2ikm} \\ & + 0.2L_{1ik,m-1} + 0.15L_{2ik,m-1} + \beta_{\text{trt}}\mathbb{I}(S_{ik} \leq \tau, m \geq S_{ik}). \end{aligned} \quad (2)$$

Treatment affects the event hazard only if initiation occurs within the grace period and the current interval is at or after initiation. Initiation after τ does not activate the treatment effect. There is no direct site effect in Equation (2); ArraySim5-3 isolates site heterogeneity in treatment initiation rather than site heterogeneity in outcome risk.

Within each interval, the simulation first generates the current L values, then treatment initiation, and finally the terminal event. A patient may therefore initiate and experience the event in the same interval, with treatment initiation occurring first.

4 CCW strategies and artificial censoring

Each patient is conceptually cloned into two strategies:

- $g = 1$: initiate treatment within the grace period,
- $g = 0$: do not initiate treatment within the grace period.

The code implements these clones through strategy-specific artificial censoring times rather than by physically duplicating the data.

For $g = 1$, a patient who fails to initiate by τ deviates during interval τ and is censored at the end of interval $\tau - 1$:

$$C_{ik}^1 = \begin{cases} M + 1, & S_{ik} \leq \tau, \\ \tau - 1, & S_{ik} > \tau. \end{cases}$$

For $g = 0$, a patient who initiates during interval $S_{ik} \leq \tau$ deviates during that interval and is censored at the end of the preceding interval:

$$C_{ik}^0 = \begin{cases} S_{ik} - 1, & S_{ik} \leq \tau, \\ M + 1, & S_{ik} > \tau. \end{cases}$$

This convention excludes an event in the initiation interval from the $g = 0$ clone but allows it to contribute to the $g = 1$ clone.

5 Oracle truth

5.1 Oracle population

For every combination of $(\tau, \beta_{\text{trt}}, h)$, an oracle cohort of $N_{\text{oracle}} = 3,000,000$ patients is simulated. The cohort is allocated equally across the three sites, so the target population has site weights

$$\pi_k = \frac{1}{3}, \quad k = 1, 2, 3.$$

The oracle uses the scenario-specific site intercepts and the same DGP as the finite samples. Common random-number seeds are used across grid cells to reduce unnecessary Monte Carlo variation.

5.2 True denominator and common numerator

The oracle denominator is the true treatment-initiation probability from Equation (1), rather than a fitted regression:

$$q_{ikm}^{D,\text{true}} = \text{expit}(\alpha_{kh} + 0.8X_{1ik} - 0.3X_{2ik} + 0.5L_{1ikm} + 0.4L_{2ikm}).$$

The common numerator hazard in the oracle population is

$$q_m^{N,\text{oracle}} = \frac{\sum_{k=1}^3 \sum_i \mathbb{I}(S_{ik} = m, T_{ik} \geq m)}{\sum_{k=1}^3 \sum_i \mathbb{I}(S_{ik} \geq m, T_{ik} \geq m)}. \quad (3)$$

Because treatment initiation is generated before the event within an interval, a patient with $T_{ik} = m$ is eligible for the initiation decision at interval m .

5.3 Oracle weights

Write $q_m^N = q_m^{N,\text{oracle}}$ and $q_{ikm}^D = q_{ikm}^{D,\text{true}}$. Before initiation, the incremental waiting factor is

$$r_{ikm}^{\text{wait}} = \frac{1 - q_m^N}{1 - q_{ikm}^D},$$

and at the initiation interval the incremental initiation factor is

$$r_{ikm}^{\text{init}} = \frac{q_m^N}{q_{ikm}^D}.$$

For the $g = 1$ clone, the stabilized weight accumulates waiting factors before initiation and an initiation factor at S_{ik} :

$$W_{ikm}^1 = \prod_{j=1}^{\min(m,\tau)} \left\{ \left(\frac{q_j^N}{q_{ikj}^D} \right)^{\mathbb{I}(S_{ik}=j)} \left(\frac{1 - q_j^N}{1 - q_{ikj}^D} \right)^{\mathbb{I}(S_{ik}>j)} \right\}, \quad (4)$$

where factors are evaluated only while the patient remains alive and eligible for initiation. For the $g = 0$ clone,

$$W_{ikm}^0 = \prod_{j=1}^{\min(m,\tau,S_{ik}-1)} \frac{1 - q_j^N}{1 - q_{ikj}^D}. \quad (5)$$

After τ , weights are carried forward without further updating. The current simulation uses quantile limits $(0, 1)$, which results in no effective weight truncation.

5.4 Oracle survival and target quantities

For strategy g , define

$$R_{ikm}^g = \mathbb{I}(T_{ik} > m - 1, C_{ik}^g > m - 1)$$

and

$$D_{ikm}^g = \mathbb{I}(R_{ikm}^g = 1, T_{ik} = m, T_{ik} \leq C_{ik}^g).$$

The oracle weighted hazard and survival curve are

$$\lambda_m^{g,\text{oracle}} = \frac{\sum_{k,i} W_{ikm}^g D_{ikm}^g}{\sum_{k,i} W_{ikm}^g R_{ikm}^g}, \quad S^{g,\text{oracle}}(m) = \prod_{j=1}^m \left(1 - \lambda_j^{g,\text{oracle}} \right).$$

The oracle therefore defines the Monte Carlo truth for the CCW estimand under the equal-site target population and pooled marginal initiation-timing distribution represented by the common numerator.

At $M = 25$, define

$$\psi_g = 1 - S^g(M), \quad \text{RMST}_g = \sum_{m=0}^{M-1} S^g(m).$$

The reported contrasts are

$$\begin{aligned} \text{RD} &= \psi_1 - \psi_0, & \text{RR} &= \frac{\psi_1}{\psi_0}, \\ \text{OR} &= \frac{\psi_1/(1 - \psi_1)}{\psi_0/(1 - \psi_0)}, & \text{RMST}_{\text{diff}} &= \text{RMST}_1 - \text{RMST}_0. \end{aligned}$$

6 Federated CCW

6.1 Site-specific denominator nuisance models

Each site constructs a person-interval dataset containing intervals in which the patient is alive and has not initiated before the interval. Site k fits the logistic model

$$\begin{aligned} \text{logit}(\hat{q}_{ikm}^D) &= \hat{\gamma}_{km} + \hat{\beta}_{1k}X_{1ik} + \hat{\beta}_{2k}X_{2ik} + \hat{\beta}_{3k}L_{1ikm} + \hat{\beta}_{4k}L_{2ikm} \\ &\quad + \hat{\beta}_{5k}L_{1ik,m-1} + \hat{\beta}_{6k}L_{2ik,m-1}. \end{aligned} \tag{6}$$

The interval effect $\hat{\gamma}_{km}$ is represented by a factor for m . Equation (6) is more flexible than the initiation DGP: it allows a separate baseline initiation hazard by interval and includes lagged L terms whose coefficients are zero in the true initiation model.

All coefficients in Equation (6) are estimated separately within each site. There is no cross-site sharing or partial pooling of the denominator regression coefficients.

6.2 Federated common numerator

Site k releases the interval-level counts

$$N_{km}^{\text{init}} = \sum_i \mathbb{I}(S_{ik} = m, T_{ik} \geq m)$$

and

$$N_{km}^{\text{eligible}} = \sum_i \mathbb{I}(S_{ik} \geq m, T_{ik} \geq m).$$

The central server forms

$$\hat{q}_m^N = \frac{\sum_k N_{km}^{\text{init}}}{\sum_k N_{km}^{\text{eligible}}} \tag{7}$$

and broadcasts this common numerator vector to all sites. Each site then constructs Equations (4) and (5), replacing the oracle quantities with $\hat{q}_m^N$ and $\hat{q}_{ikm}^D$.

6.3 Local weighted sufficient statistics

Using its local individual-level records, site k computes

$$DW_{km}^g = \sum_i W_{ikm}^g D_{ikm}^g, \quad RW_{km}^g = \sum_i W_{ikm}^g R_{ikm}^g.$$

Only these weighted event and risk totals are required centrally for point estimation. The central server calculates

$$\hat{\lambda}_m^{g,\text{fed}} = \frac{\sum_k DW_{km}^g}{\sum_k RW_{km}^g} \quad (8)$$

and

$$\hat{S}^{g,\text{fed}}(m) = \prod_{j=1}^m \left(1 - \hat{\lambda}_j^{g,\text{fed}}\right).$$

The treatment-effect estimates are obtained by substituting these survival curves into the RD, RR, OR, and RMST formulas above. This is a pooled weighted survival estimator: site-specific effect estimates are not calculated and then meta-analyzed.

6.4 Federated variance calculation

After computing the global hazards and survival curves, the central server returns the quantities needed for a second local pass. A strategy-specific incremental influence contribution is

$$U_{ikm}^g = \frac{1}{1 - \hat{\lambda}_m^g} \frac{W_{ikm}^g}{\sum_k RW_{km}^g} \left(D_{ikm}^g - \hat{\lambda}_m^g R_{ikm}^g \right).$$

Sites calculate individual influence-function contributions and release sums of their squares. For example,

$$\widehat{\text{SE}}(\widehat{\text{RD}}) = \left\{ \sum_k \sum_i \left(IF_{ik}^{\psi_1} - IF_{ik}^{\psi_0} \right)^2 \right\}^{1/2}.$$

Analogous calculations are used for log-RR, log-OR, and RMST contrasts. The implemented variance calculation treats the estimated weights as fixed and does not add nuisance-model estimation uncertainty.

7 Aligned fully site-stratified pooled CCW

The pooled comparator has access to all individual-level records at a central location, but it deliberately uses the same statistical specification as the federated estimator. Specifically, it

1. separates the pooled data by site;
2. fits Equation (6) separately within every site;
3. constructs the same common numerator in Equation (7);
4. applies the same artificial censoring and weight construction;
5. pools the weighted outcome records; and
6. applies the same survival and influence-function calculations.

It is not a conventional pooled logistic model with a single common slope vector and site fixed effects. It is a fully site-stratified individual-data implementation of the same estimator used by Fed-CCW.

The pooled weighted hazard is

$$\hat{\lambda}_m^{g,\text{pool}} = \frac{\sum_k \sum_i W_{ikm}^g D_{ikm}^g}{\sum_k \sum_i W_{ikm}^g R_{ikm}^g}. \quad (9)$$

Because

$$DW_{km}^g = \sum_i W_{ikm}^g D_{ikm}^g, \quad RW_{km}^g = \sum_i W_{ikm}^g R_{ikm}^g,$$

Equations (8) and (9) imply

$$\hat{\lambda}_m^{g,\text{fed}} = \hat{\lambda}_m^{g,\text{pool}}.$$

It follows that

$$\hat{S}^{g,\text{fed}}(m) = \hat{S}^{g,\text{pool}}(m)$$

and hence

$$\hat{\theta}^{\text{fed}} = \hat{\theta}^{\text{pool}}$$

for every reported CCW estimand θ . Because the pooled comparator also uses the same influence-function decomposition, the implemented model-based standard errors agree up to numerical precision.

The aligned pooled CCW is therefore an implementation gold standard rather than a statistically different competing estimator. Its purpose is to show that the federated exchange of common numerator counts, weighted event/risk totals, and influence-function summaries reproduces the corresponding individual-data analysis without transferring individual records.

8 Where information is and is not shared

The denominator nuisance regressions are fitted independently, so no site borrows regression coefficients from another site. Cross-site information is combined in two places:

1. the common numerator $\hat{q}_m^N$, estimated from aggregated initiation and eligibility counts; and
2. the final outcome estimator, formed by aggregating weighted event, risk, and influence-function summaries.

Consequently, ArraySim5-3 evaluates lossless federation for a fully site-stratified nuisance-model specification. It does not evaluate a shared slope, random-effect, or hierarchical nuisance model.

9 Monte Carlo performance evaluation

For each replicate r and estimand θ , performance is evaluated against the cell-specific oracle truth:

$$\begin{aligned} \text{Error}_r(\theta) &= \hat{\theta}_r - \theta_{\text{oracle}}, \\ \text{Bias}(\theta) &= \frac{1}{R} \sum_{r=1}^R \left(\hat{\theta}_r - \theta_{\text{oracle}} \right), \end{aligned}$$

and

$$\text{RMSE}(\theta) = \left\{ \frac{1}{R} \sum_{r=1}^R \left(\hat{\theta}_r - \theta_{\text{oracle}} \right)^2 \right\}^{1/2}, \quad R = 100.$$

Because Fed-CCW and the aligned pooled CCW implement the same estimator, their replicate-level estimates, biases, RMSEs, and current model-based confidence intervals are expected to overlap.